PSYC4411

CHALLENGE LAB: VISUALISING LIFESTYLE & MENTAL HEALTH

Week 2 · Semester 1, 2026

Current Advances in Psychological Methods & Analyses

🎯 TODAY'S CHALLENGE

- Create a **multi-panel visualisation** exploring lifestyle factors and mental health in a synthetic dataset (3,000 participants, 44 variables including DASS-21 scores)
  - Sleep, exercise, screen time, social media, caffeine, income — which matter most?
- Try **two workflows**: Jupyter notebook and Python scripts
  - Neither is "better" — the point is to try both and form a preference
- Apply the **LLM Problem-Solving Loop** from Week 1
  - Plan first → then code → then verify → then document
- Prepare **1 HTML slide** (~3 min) for next week

YOUR AI TOOLKIT

Use **whatever AI tools you like**. No rules, no restrictions. Here are some options:

INSIDE VS CODE

- **Chat** — ask questions, get code snippets, paste them in
- **Edit** — highlight code, ask the AI to modify it in place
- **Agent** — let the AI create files, run code, and iterate autonomously

OUTSIDE VS CODE

- **Browser chatbots** — ChatGPT, Claude, Gemini — copy-paste code back and forth
- **CLI tools** — Claude Code, Codex — if you've installed them, go for it
- **Mix and match** — use different tools for different tasks

WHY LEARN VS CODE?

- Browser chatbots are great for questions and quick code — but you still have to **copy, paste, run, debug** yourself
- VS Code's AI features can **edit your files directly**, run your code, read error messages, and fix problems — a tighter loop
- Over this semester you'll see how this scales:

  > **LLM → Code → Solution**
  > Having an AI *write and run code* to produce a figure, analysis, or document is more powerful and reproducible than asking an AI to generate those outputs directly

- No pressure today — use whatever feels comfortable. But **getting familiar with VS Code now pays off later**

> **VS Code Coding Assistants**
> You can install extensions for **Claude Code**, **Codex** (OpenAI), and **Gemini** directly inside VS Code — each gives you Chat, Edit, and Agent modes. Gemini works with any Gmail account; Codex and Gemini both offer a **free tier**. If you hit the limit on one, just switch to another — they all work the same way.

🚀 GETTING STARTED

Pick **one** workflow, or try **both** to see which you prefer — there's no wrong choice.

OPTION A: NOTEBOOK

1. Open `starter.ipynb`
2. Run the setup cells (imports, data loading)
3. **Ask the AI to plan** before writing code
4. Build your visualisation cell by cell
5. Verify, refine, document

OPTION B: SCRIPTS

1. Ask the AI to create a `plan.md`
2. Review and iterate on the plan
3. Have the AI create a `.py` script
4. Run it: `python your_script.py`
5. Check the PNG, refine, document

Full details in the challenge brief: `weeks/week-02-lab/README.md`